

Prediction of Prices in Agriculture (PREPARE)

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Project Overview

The main idea behind our project is to solve farmer's price blindness. Right now, small and marginal farmers don't have accurate, timely price forecasts to understand market dynamics. We also want to provide reliable multi-day price prediction tools so that the government and regulators can plan better market interventions.

Please refer to the previous reports for semester 5 progress and progress made in semester 6 prior to the current report.

Progress Since the Last Report

Following our previous imputation benchmarking, we ran thresholding tests on all the 10 imputation strategies targeting the 50% to 75% data availability range (in 5 percent intervals). During this analysis, we figured out that DOW Ratio was still the best overall technique across this range by RMSE (save at the 75% interval where the rolling mean strategy worked best). We also found out that most mandis are physically closed on Sundays, yet they often still show recorded data for those days. Conversely, data is frequently missing for operational weekdays. This appears to stem from administrative filing discrepancies and the practice of uploading mandi data retroactively.

Using our finalized dataset (imputed using the DOW Ratio for mandis with at least 50% data availability), we experimented with several model architectures to extend our prediction horizon. We trained and evaluated HistGradientBoostingRegressor and STGCN models. While these architectures yielded solid results for single-day predictions, their performance dropped significantly when forecasting over a 15-day period. Interestingly, the model predicting potato prices maintained largely accurate forecasts over the extended horizon, but the models for onion, tomato, and wheat struggled to achieve the same stability.

During feature engineering, we noticed that incorporating the prices of other crops as features improved the target model's performance (for instance, feeding tomato prices as an input column into the onion prediction model). We discussed this finding in our latest meeting with Prof. Bhardwaj. We hypothesized that this cross-crop correlation could occur because certain crops act as leading indicators, reacting earlier to broader market events or supply chain disruptions. This is a working theory that we need to experiment with further to confirm. We also discussed other possible experiments such as subtracting a fixed baseline from all prices or zero centering the data entirely, alongside trying new normalization techniques for our other variables. We also discussed exploring alternative models such as the SAGMM (Self-Adaptive Graph Mixture of Models) architecture as a potential upgrade.

Next Steps

Over the next few weeks, our plan is to:

- Experiment with shifting the price values to improve model accuracy. This includes subtracting a fixed baseline from all prices (if prices range from 320-390, subtracting 300 to narrow the scale) or by subtracting the mean.
- Explore and apply different normalization techniques across our other input variables.
- Establish a baseline by duplicating the previous day's price as the current day's prediction and calculate the R-squared for this to benchmark our models.
- Research and explore alternative architectures, specifically looking into integrating SAGMM (Self-Adaptive Graph Mixture of Models) and Graph WaveNet.
- Focus on model improvement, developing the architecture for higher multi-day prediction horizons (1-15 days).
- Group the mandis into clusters based on their production volume and test if this clustering affects the model's overall efficacy.